INCH-POUND

KKK-T-640E
5 DECEMBER 1990
SUPERSEDING
KKK-T-640D
June 29, 1975

FEDERAL SPECIFICATION

TRACTOR, WHEELED, AGRICULTURAL, (GASOLINE-ENGINE DRIVEN, DIESEL-ENGINE DRIVEN OR LP-GAS ENGINE DRIVEN)

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers agricultural wheeled tractors developed for use as a mobile unit for performing functions requiring one or more of the following applications of power: (1) drawbar, (2) power takeoff, or (3) hydraulic.

1.1.1 Federal specification coverage. Federal specifications do not include all types and sizes of the commodity indicated by the titles of the specification, but cover only those generally used by the Government.

1.2 Classification.

1.2.1 Sizes. The tractors covered by this specification shall be of the following minimum sizes (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2420

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TABLE I. Tractor sizes and minimum drawbar horsepower.

Tractor size	Minimum drawbar horsepower*
W-20	20
W-25	25
W-30	30
W-35	35
W-40	40
W-45	45
W-50	50
W-55	55
W-60	60
W-65	65
W-70	70
W-75	75
W-80	80
W-85	85
W-90	90
W-95	95
W-100	100
W-110	110
W-120	120
W-130	130
W-140	140
W-150	150
W-175	175
W-200	200

* To determine the minimum drawbar horsepower in qualifying for the appropriate tractor size, the maximum drawbar horsepower with ballast, based on the most recent Nebraska test data, shall be used.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Federal Specifications:

F-F-351 - Filters and Filter Elements, Fluid, Pressure: Lubricating Oil, Bypass and Full Flow.

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O-E-915 - Extinguishers, Fire, Dry-Chemical, (Portable).
ZZ-T-1619 - Tires, Pneumatic, Agricultural.

Federal Standard:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, and standards as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY* Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Auburn, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-T-3351 - Tractor, Full Tracked, Low Speed, Tractor, Wheeled, Agriculture and Tractor, Wheeled, Industrial, and Their Attachments, Packaging of.

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein.

Laws and Regulations:

29 CFR 1926.600 - Equipment.
29 CFR 1926.602 - Material handling equipment.

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29 CFR 1928.51 - Roll-over protective structures (ROPS) for tractors used in agricultural operations.
41 CFR 1-1.25 - Recovered Materials.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1 - Spark Arrestor for Internal Combustion Engines.

(Application for copies should be addressed to the Division of Administrative Service, Forest Service, United States Department of Agriculture, Washington, DC 20250.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF AGRICULTURAL ENGINEERS (ASAE) STANDARD

ASAE S201.4 - Application of Hydraulic Remote Control Cylinders to Farm Tractors and Trailing Type Farm Implements.

(Application for copies should be addressed to the American Society of Agricultural Engineers, 2950 Niles Road, St. Joseph, MI 49085.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Unfired Pressure Vessels (Section VIII).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL BOARD OF FIRE UNDERWRITERS' CODE

No. 58 - Liquefied Petroleum Gases.

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York, NY 10038; 222 West Adams Street, Chicago, IL 60606; 465 California Street, San Francisco, CA 94104.)

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THE NEBRASKA TRACTOR TEST LAW

(Application for copies should be addressed to the Nebraska Tractor Test Laboratory, East Campus, Lincoln, Nebraska 68503.)

THE TIRE AND RIM ASSOCIATION, INC. YEARBOOK

(Application for copies should be addressed to The Tire and Rim Association, Inc. 3200 West Market Street, Akron, OH 44313.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J88	-	Exterior Sound Level Measurement for Powered Mobile Construction Equipment.
J137	-	Lighting and Marking of Agricultural Equipment on Highways.
J154	-	Access Systems for Off-Road Machines.
J185	-	Access Systems for Off-Road Machines.
J208	-	Safety for Agricultural Equipment.
J209	-	Instrument Face Design and Location for Construction and Mechanical Equipment.
J284	-	Safety Alert Symbol for Agricultural, Construction, and Industrial Equipment.
J350	-	Spark Arrestor Test Procedure for Medium Size Engines.
J534	-	Lubrication Fittings.
J560	-	Seven Conductor Electrical Conductor for Truck Trailer Jumper Cable.
J712	-	Tractor and Implement Disc Wheels.
J715	-	Three Point Free Link Hitch Attachment of Implements to Agricultural Wheeled Tractors.
J716	-	Application of Hydraulic Remote Control to Agricultural Tractors and Trailing type Agricultural Implements.
J718	-	540-RPM Power Take-Off for Farm Tractors.
J719	-	1000-RPM Power Take-Off for Farm Tractors.
J725	-	Mounting Brackets and Socket for Warning Lamp and Slow-Moving Vehicle (SMV) Identification Emblem.
J833	-	USA Human Physical Dimensions.
J841	-	Operator Controls for Agricultural Wheeled Tractors.
J898	-	Control Locations for Off-Road Work Machines.
J919	-	Sound Level Measurements at the Operator's Station for Agricultural and Construction Equipment.
J925	-	Minimum Service Access Dimensions for Off-Road Machines.
J974	-	Flashing Warning Lamp for Agricultural Equipment.
J975	-	Headlamps for Agricultural Equipment.
J976	-	Combination Tail and Floodlamp for Agricultural Equipment.
J1362	-	Symbols for Controls, Indicators, and Tell-Tales for Off-Road, Self-Propelled Work Machines.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

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3. REQUIREMENTS

3.1 General. This specification covers agricultural wheeled tractors, complete with necessary operating accessories customarily furnished with tractors of this type whether stipulated herein or not, together with such modifications and attachments as may be necessary to enable the unit to function reliably and efficiently in sustained operation.

3.2 Tractor type. The tractor types shall be as specified herein (see 6.2).

3.2.1 Standard type (utility). Standard-type tractors shall be of conventional four-wheel design with both front and rear tread width fixed or adjustable.

3.2.2 Wide adjustable front and rear axle row-crop type. The adjustable front and rear-axle type tractors shall have both front and rear treads adjustable. Clearance, row requirements, and tread adjustment within the commercial range shall be as specified by the contracting officer (see 6.2).

3.2.3 High clearance row-crop type. Unless otherwise specified, tread requirements and cultivating clearance shall be the manufacturer's current high clearance, cane or rice field type tractor.

3.3 First article. When specified (see 6.2), a tractor shall be subjected to first article inspection (see 4.3 and 6.2).

3.3.1 Material. Material shall be as specified herein. Material not specified shall be the manufacturer's current as offered to the public differing only in respects as necessary to meet special requirements. Components of electrical, fuel, and exhaust systems shall resist the harmful effects of dust, moisture, and corrosion.

3.3.2 Material deterioration prevention and control. The tractor shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the tractor may be exposed.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents and shall make information available, upon request, to the contracting officer or designated representative.

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3.3.5 Recovered materials. The contractor/offeror is encouraged to use recovered materials in accordance with Public Law 94-580 (41 CFR 1-1.25) to the maximum extent practical. "This shall not be interpreted to mean that the use of used or rebuilt products is allowed under this document."

3.4 Operational requirements. The tractor shall be capable of starting and operating efficiently under any of the conditions or combination of conditions outlined below.

3.4.1 Temperature conditions. Without preheating from an external source the engine shall start within 5 minutes, and the engine shall operate at not less than 80 percent of the applicable Nebraska full-rated PTO (Power-Take-Off) horsepower at rated engine speed within 30 minutes after engine start in any ambient temperature from -15 to +110 °F.

3.4.2 Use conditions. Design and construction shall be such that the tractor will withstand usage encountered in normal agricultural service for extended periods of time. Components, particularly of the electrical and fuel systems, shall be so designed and protected as to resist any harmful effects of dust or water. This shall include a rain cap on a vertical exhaust or the manufacturer's standard curved piping as regular equipment.

3.5 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect mitten compatible fastenings where practicable. Drain outlets shall be located for accessibility and shall permit the use of receptacles for collecting drainage.

3.6 Frame. The frame or chassis shall withstand the maximum stresses encountered in normal operating conditions. It shall be capable of providing support for attaching farm implements or tools designed as mounted equipment and approved by the tractor manufacturer for use in combination with the tractor.

3.7 Engine. The engine shall be as specified herein (see 6.2).

3.7.1 Gasoline engine. When specified, the engine shall be an internal combustion engine of the four-stroke cycle, of two or more cylinders, complete with all necessary operating accessories and capable of satisfactory operation on gasoline as recommended by the manufacturer. The engine shall be equipped with high-tension ignition system, dry-filter type air cleaner, efficient lubricating oil system, oil filtering system and exhaust system.

3.7.2 Diesel engine. When specified, the engine shall be of the compression ignition type, either two or four-stroke cycle, capable of operating on diesel or JP-8 fuel. The engine shall be equipped with an oil-filter system, efficient lubricating oil system, air cleaner of the dry-filter type and exhaust system. When specified (see 6.2) pre-heater using an external power source shall be furnished.

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3.7.3 Liquefied petroleum (LP) gas engine. When specified, the engine shall be an internal combustion engine of the four-stroke-cycle, liquid-cooled type of two or more cylinders, capable of efficient operation on any mixture of commercially available LP gas fuels as recommended by the manufacturer. The engine shall be equipped with a liquid fuel withdrawal system including filter regulator, vaporizer, and carburetor especially designed for use with these fuels. Fittings, lines, and components shall be in accordance with National Board of Fire Underwriters' Code No. 58 for Liquefied Petroleum Gases for maximum safety for the equipment and operating personnel. The engine shall be equipped with a high-tension ignition system, dry-filter type air cleaner, lubricating oils system oil filtering system and exhaust system.

3.7.4 Oil filter. A full-flow type oil filter shall be installed on the pressure side of the engine oil pump in an accessible location. The oil filter shall consist of a metal container and a replaceable element or a one-piece disposable type so constructed that no special equipment or skill will be required in changing the filtering elements. The filtering elements shall be enclosed in such a manner that the filter material shall be prevented from leaving the filter and passing into other parts of the oil-circulating system. A bypass type oil filter, when provided, shall have an efficiency as required by test in accordance with when provided F-F-351. The full-flow type oil filter element must meet or exceed the engine manufacturer's lubrication requirements as to efficiency and flow.

3.7.5 Engine governor. The engine governor shall be of the variable-speed centrifugal type and shall be gear driven or shaft driven from the engine. A hand governor control for regulating the governed speed setting throughout the entire engine load range shall be provided and mounted so that it will be convenient to the operator. When specified (see 6.2), an overriding governor-type foot control shall be provided.

3.7.6 Engine cooling system. Unless otherwise specified (see 6.2), the contractor's current standard production engine cooling system will be acceptable.

3.7.7 Engine starting system. The contractor's standard power starting system will be acceptable.

3.7.8 Fuel tank. The manufacturer's current standard capacity fuel tank will be acceptable and shall be protected so it is not affected by heat from the engine or exhaust system. The fuel system shall be equipped with a fuel sediment trap and shutoff valve. Adequate fuel filters shall be provided for diesel tractors. LP gas tanks shall be in accordance with ASME code for unfired pressure vessels. The type of fuel used (i.e., DIESEL ONLY, MOGAS ONLY) shall be prominently labeled near the fuel fill port. The label characters shall have a minimum height of .75 inches.

3.7.9 Lights and circuits. The following lights with associated circuits, circuitbreakers, or other protective devices and switches shall be provided: Head lamps shall conform to SAE J975; tail and floodlamp shall conform to

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SAE 976; flashing warning light shall conform to SAE J974; and light and marking shall conform to SAE J137.

3.7.10 Charging system. The alternator with integral regulator shall be the contractor's standard, adequate for use with the battery specified and with the electrical loads associated with 3.7.9, and any other accessories consuming electrical current. Alternators with integral regulators will not show discharge rate with all electrical components in the "on" position when engine is at idle.

3.7.10.1 Battery. The battery or batteries shall be the manufacturer's standard of the type regularly furnished for cold starting, as specified (see 3.6). Mounting shall permit ready accessibility for servicing, removing, and replacement.

3.8 Clutch. The clutch, torque converter, or hydrostatic drive, as specified (see 6.2), shall be the contractor's current standard type.

3.9 Transmission. The transmission shall be of either the selective sliding-gear, synchromesh, constant-mesh planetary type, or hydraulic type, except as specified herein. The input torque capacity of the transmission shall be greater than the maximum gross torque developed by the engine. Operation of the gear-type transmission augmented by planetary gear or mechanical sliding-gear units is acceptable. Either torque amplifier, torque converter of shuttle type, hydrostatic or planetary-type drive may be specified in lieu of the above (see 6.2).

3.9.1 Number of transmission speeds. The contractor's standard transmission speeds for size tractor specified shall be furnished.

3.10 Steering. When specified (see 6.2), power-assist or full power steering shall be furnished. The power steering mechanism shall preserve the features of directional control inherent in the manually operated steering mechanism. While the engine is at idle and the tractor is stationary on a hard surface, the force required by the operator to turn the steering wheel from the extreme counterclockwise position to the extreme clockwise position and back, shall not exceed 15 pounds when measured at the wheel rim. If the tractor is equipped with power assisted steering, the force required to turn the steering wheel from the extreme counterclockwise position to the extreme clockwise position and back, with the power assist mechanism inoperative and the tractor sitting stationary on a hard surface, shall not exceed 50 pounds when measured at the wheel rim.

3.11 Final drive units. The final drive units shall operate in fully enclosed housing in constant oil bath.

3.12 Wheels and rims. Steel disc wheels shall conform to SAE J712. In lieu thereof, cast or welded steel wheels are permissible. Unless otherwise specified (see 6.2), rims shall be in accordance with the Tire and Rim Association Yearbook for the tire size required. The recommended tire

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inflation pressure, as described in the TRA yearbook, shall be stencilled near each wheel.

3.13 Tires. Contractor's standard agricultural tires for size tractor offered will be furnished unless special size or type tires are required (see 6.2). Tire and Rim Manufacturer's Association rating tables shall be used to determine minimum compliance with this specification. Tires furnished shall meet the minimum requirements of ZZ-T-1619.

3.14 Brakes. The tractor shall be provided with brakes which shall control the tractor. The brakes shall be of hydraulic, mechanical, with or without power-assist type, designed to operate either the right or left brake independently or simultaneously at the discretion of the operator. The brakes shall permit control of the tractor under all normal operating conditions and be acceptable for deceleration as well as steering purposes. A mechanical parking brake, which is part of the main braking system, or an independent mechanical parking device shall be provided. Mechanical service brakes that can be locked as parking brakes are acceptable. Transmission parking locks in lieu of parking brakes are acceptable only if the drive axle is equipped with a differential lock and a prominently displayed warning is permanently attached to the instrument panel stating that the differential lock must be engaged when parking on slippery surfaces. While pulling a rated load up a 30 percent, loose soil grade, the force that the operator must exert on the brake pedals to stop the vehicle, disengage the transmission, and hold the tractor and load stationary (with transmission disengaged) shall not exceed 40 pounds.

3.15 Rear fenders. The tractor shall be equipped with the contractor's current standard production rear fenders or guards.

3.16 Operator's seat. Unless otherwise specified (see 6.2), the tractor shall be equipped with the contractor's current standard production seat.

3.17 Power lift. Unless otherwise specified (see 6.2), the tractor shall be equipped with a three-point free link hitch conforming to SAE J715. When specified (see 6.2), a quick-attaching coupler and a remote control for trailing type implements, conforming to SAE J716 will be included.

3.18 Drawbar hitch. The tractor shall be equipped with or without a hitch, as specified. Unless otherwise specified (see 6.2), the type of drawbar hitch is one or more of the following:

a. Swinging drawbar type.
b. Fixed drawbar type.
c. One-point hitch.
d. Three-point lift type hitch. (A removable bar between lower link.

In the event a three-point lift type hitch is specified, it shall conform to requirements set forth in SAE J715.

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3.19 Power takeoff. When specified (see 6.2), any tractor, 20-drawbar horsepower and above, shall be equipped, with rear power takeoff connection in conforming to SAE J718 and SAE J719, as applicable.

3.20 Instruments. The following instruments shall be provided, so mounted as to be convenient and visible from the operator's seat:

Oil-pressure gauge
Engine temperature gauge
Ammeter or voltage gauge
Fuel gauge
Light switch
Starter control
Ignition switch

Key operated ignition switch shall be furnished. Hand choke shall be furnished on gasoline models. All instruments shall be in accordance with SAE J209.

3.21 Tachometer and engine hour meter. The tractor shall be equipped with a tachometer and hour meter or combination thereof, which shall register the number of hours of operation of the engine. The totalizing mechanism shall register not less than 9,999 hours of operation. The tachometer and hour meter shall be the standard product regularly furnished by the contractor.

3.22 SMV emblem. Unless otherwise specified (see 6.2), a slow moving vehicle (SMV) emblem conforming to SAE J943 shall be provided. The mounting bracket and socket for the warning lamp shall conform to SAE J725.

3.23 Lubrication. All moving parts requiring lubrication shall be provided with suitable means of lubrication and located in accessible, protected positions. Fittings shall comply with SAE J534.

3.24 Exhaust system. The engine shall be equipped with the contractor's current standard exhaust system. An effective device shall be furnished, if necessary, to prevent the entry of rain into the exhaust system.

3.25 Spark arrestor. When specified (see 6.2), the tractor shall be equipped with a spark arrestor in accordance with Forest Service Standard 5100-1A. The spark arrestor and muffler may be a combined unit.

3.26 Hydraulic system. When specified (see 6.2), the contractor's standard hydraulic system shall be furnished on the tractor. The system shall conform to the use conditions in 3.4.2 and intended use in 6.1.

3.27 Tools. Tools normally furnished with the tractor by the contractor shall be provided together with the contractor's regular tool storage space.

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3.28 Painting. Unless otherwise specified (see 6.2), all surfaces exposed to weather, other than corrosion-resistant polished surfaces or trim, shall be painted using the contractor's standard painting procedure and color.

3.29 Manufacturer's identification. Each tractor shall bear the manufacturer's name or trademark, model, and serial number either cast integrally with, stamped, or marked on the tractor or on a corrosion-resistant metal plate securely affixed to the tractor.

3.30 Implements, allied equipment, and attachments. When implements, allied equipment, and attachments are specified (see 6.2), such equipment shall be a current model, so designed and constructed to be readily adaptable without modification by the procuring agency, and shall operate efficiently with the tractors so equipped.

3.31 Optional accessories and allied equipment. When optional accessories and allied equipment are specified and furnished, such equipment shall be of current models so designed and constructed as to be readily adaptable without modification, and shall operate with the tractors so equipped. When specified (see 6.2), tractors shall be furnished with any of the following, or other accessories as specified by the contracting officer.

3.31.1 Optional accessories.

a. Power take-off:

(1) Transmission-driven power take-off, 540 rpm.
(2) Continuous-running power take-off, specify 540 or 1,000 rpm.
(3) Independent-power take-off, specify 540 or 1,000 rpm or combination thereof.

b. Power lift:

(1) For mounted equipment.
(2) For other equipment with remote control system as specified, conforming to ASAE 201.4.

c. Additional weights (specify kind and minimum weight):

(1) Front wheel (pair).
(2) Rear wheel (pair).
(3) Frame weights (front).
(4) Frame weights (rear).

d. Fire extinguisher conforming to O-E-915, size 2-1/2, mounted in a position readily accessible to the operator.
e. Mechanical or spark-plug type tire pump.

f. High altitude equipment - specify altitude.
g. Front bumper to cover the width of the frame or radiator.

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h. Locking device for tool storage space.
i. Operator's seat belt.
j. Where the buyer's needs are primarily for power take-off usage, the tender may specify that the tractor sizes in 1.2.1 shall be determined based on the most recent Nebraska tests, on the maximum power take-off horsepower at rated engine speed. The minimum size horsepower figures in 1.2.1 shall remain unchanged. When this option is specified, the term power take-off horsepower shall be substituted throughout this specification for the term drawbar horsepower.
k. A differential lock.
l. Cab, fully enclosed protective type. The cab shall be the contractor's current standard production type, enclosing the operator's seat and all controls. Power operated windshield wipers shall be provided; one front and one rear. If cab has a separated or two-piece windshield, two power operated front wipers shall be required. Cab shall be provided with the contractor's standard rollover protection. The cab shall be provided with opening doors or either side.
m. Personnel heater, defroster or defroster fan installed in fully enclosed cab.
n. Air conditioner installed in fully enclosed cab.

3.32 Workmanship. Workmanship shall conform to current best manufacturing practice followed for this type of equipment. Component parts and units shall be manufactured to definite standard dimensions with proper fits and clearance. All parts shall be clean, free from rust, toolmarks, pits and other injurious defects. External surfaces shall be free of burrs, sharp edges, and corners except where sharp edges or corners are required and are not detrimental to safety.

3.33 Safety. All exposed parts subject to high operating temperatures or which are energized electrically shall be located, insulated, fully enclosed, or guarded to prevent hazard to operating personnel. All moving parts which are of such nature or so located as to be a hazard to operating personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions of the tractor. When enclosed cab is specified, safety glass or equivalent shall be provided. A horn shall also be provided.

3.34 Human factors engineering. The tractor's design shall incorporate accepted human factors engineering design criteria and practice. As a minimum, the design shall conform to the following SAE requirements: J154a, J185, J208d, J209, J284, J833, J841, J925, and J1362.

3.34.1 Noise limits. The noise produced by the tractor shall conform with MIL-STD-1474 requirements, with the exception of MIL-STD-1474, 5.2 and 5.4, when tested in accordance with 4.5.3. The provisions of MIL-STD-1474, 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform to MIL-STD-1474, 4.3, except the sign

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shall state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign.

3.35 Performance.

3.35.1 Drawbar horsepower. The tractor shall meet the minimum drawbar horsepower requirements for tractor size as established in 1.2.1, table I.

3.35.2 Gradeability. The tractor shall negotiate a combination of longitudinal and side slopes up to 30 percent on loose soil as encountered in normal agricultural operations without any wheel losing ground contact. The engine and other pressure operated and lubricated components shall maintain constant operating oil pressure and operate without malfunction under these slope conditions.

3.35.3 Operating and functional evaluation. The tractor engine and electrical system along with any of the selected accessories and installed attachments shall be capable of completing an operating and functional evaluation to determine adequacy of production assemblage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection for preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be subjected to the tests marked "x" in column 1 of table II in the order shown. Failure of any test shall be cause for rejection of the first article.

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TABLE II. Test Schedule.

First article	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Sound level.	4.5.3	3.34
X	-	Drawbar horsepower.	4.5.4	3.35.1
X	-	Gradeability.	4.5.5	3.35.2
X	X	Operation/functional.	4.5.6	3.35.3

4.4 Quality conformance inspection.

4.4.1 Examination. Each tractor shall be examined for the defects specified in 4.5.1. Presence of any defects shall be cause for rejection of the tractor.

4.4.2 Tests. Each tractor shall be subjected to tests marked "x" in column 2 of table II. Failure of any test shall be cause for rejection of the tractor.

4.5 Inspection procedure.

4.5.1 Examination. The tractor shall be examined for the following defects:

Number	Defects	Requirement paragraph
	Major	
101.	Materials not as specified.	3.3.1
102.	Recovered materials not in accordance with Public Law 94-580 (41CFR 1-1.25).	3.3.5
103.	Engine not as specified.	3.7
104.	Engine oil filter not as specified.	3.7.4
105.	Engine governor not as specified.	3.7.5
106.	Fuel tank not as specified.	3.7.8
107.	Clutch not as specified.	3.8
108.	Transmission not as specified.	3.9
109.	Steering not as specified.	3.10
110.	Wheels and rims not as specified.	3.12
111.	Tires not as specified.	3.13
112.	Brakes not as specified.	3.14
113.	Power lift not as specified.	3.17
114.	Hitch not as specified.	3.18

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115.	Power takeoff not as specified.	3.19
116.	Instruments not as specified.	3.20
117.	Emblem not as specified.	3.22
118.	Lubrication not as specified.	3.23
119.	Engine exhaust not as specified.	3.24
120.	Spark arrestor not as specified.	3.25
121.	Painting not as specified.	3.28
122.	Optional accessories not as specified.	3.31
123.	Human factors engineering not as specified.	3.34

Minor

201.	Tools not as specified.	3.27
202.	Manufacturer's identification not as specified.	3.29
203.	Workmanship not as specified.	3.32

4.5.2 Test conditions. Unless otherwise specified herein, test shall be conducted at the ambient temperature and climatic conditions existing at the place of test. Only that maintenance established by the contractor and submitted as maintenance schedule prior to commencement of the tests shall be performed during the tests.

4.5.3 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.4 Engine drawbar horsepower. When standard production units previously tested by the University of Nebraska are furnished under this specification, no drawbar horsepower test will be required for qualification. In the absence of Nebraska test reports, in order to ascertain compliance with this specification, only such test shall be prescribed as conforming to the standard test code of the University of Nebraska. A certified manufacturer's test on new untested machines shall be acceptable.

4.5.5 Gradeability. Locate the tractor on a prepared area of lose top soil similar to that prepared for normal agricultural purposes. The area should range in grade from level to slopes of at least 30 percent. The tractor shall traverse the area including the slopes both horizontally and vertically without stalling or losing traction. All tractor wheels shall remain in full ground contact while traversing any location in the area and full recommended oil pressure shall be maintained.

4.5.6 Operating and functional. The tractor shall be operated for a minimum of 2 hours. This operation shall consist of full operation of all

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electrical and hydraulic systems along with the selected attachments and accessories. The tractor while operating in the prepared area shall be subjected to a series of running turns both right and left from both forward and reverse directions coming to a complete stop between each series of turns. At least 5 engine starts shall be accomplished. Nonconformance to 3.35.3 shall constitute failure of this test.

4.5.7 Certification. The contractor may furnish certified test data to assure compliance with safety (see 3.33) and performance (see 3.35.3). Nonconformance to the requirements of 3.33 and 3.35.3 shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packing, and marking shall be examined and tested for compliance with the quality assurance provisions of MIL-T-3351 except AQL shall be 1.0 percent. Also, preservation, packing, and marking shall be examined for compliance with additional requirements specified in section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each tractor shall be preserved in accordance with the level A requirements, type II, of MIL-T-3351.

5.1.2 Level C. Each tractor shall be preserved in accordance with the level B requirements, type II, of MIL-T-3351, except strapping shall be finish A.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Each tractor, preserved as specified in 5.1, shall be packed in accordance with the level A requirements, type II, of MIL-T-3351.

5.2.2 Level C. Each tractor, preserved as specified in 5.1, shall be packed in accordance with the level B requirements, type II, of MIL-T-3351.

5.3 Marking. In addition to any special marking specified in the contract or purchase order, (see 6.2), marking shall be in accordance with MIL-T-3351, type II. For civilian agencies, the interior packages, shipping containers, semiboxed and mobile units shall be marked in accordance with FED-STD-123.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The tractor is to be used as a mobile unit for performing general agricultural-type work when equipped with optional attachments and accessories as specified for the intended purpose.

6.2 Ordering data.

a. Title, number, and date of the specification.
b. Tractor size and type (see 1.2.1 and 3.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Clearance, row requirements and tread adjustments of the wide adjustable front and rear axle row-crop type tractor (see 3.2.2).
e. When a first article is required for inspection and approval (see 3.3).
f. Type engine (see 3.7).
g. Type of pre-heater if required (see 3.7).
h. When overriding governor-type foot control (see 3.7.5).
i. Cooling system other than standard (see 3.7.6).
j. Battery other than manufacturer's standard (3.7.10).
k. Clutch, torque convertor, or hydrostatic drive (see 3.8).
l. Type of transmission (see 3.9).
m. Type steering (see 3.10).
n. Type of wheel and rims (see 3.12).
o. Tire size and tread if other than supplier's standard (see 3.13).
p. When other than supplier's standard operator's seat is required (see 3.16).
q. When power lift is required (see 3.17).
r. Type drawbar hitch required (see 3.18).
s. When power take-off is required (see 3.19 and 3.31.1).
t. When SMV emblem is required (see 3.22).
u. When spark arrestor is required (see 3.25).
v. When hydraulic system is required (see 3.26).
w. Painting, if different (see 3.28).
x. Optional accessories and attachments; specify type (see 3.31).
y. Sound-level limits (see 3.34).
z. When certified data and test reports are acceptable (see 4.5.7).
aa. Preservation and packing required (see 5.1 and 5.2).
bb. Any special marking (see 5.3).
cc. When model data and drawings are required (see 6.4.2).
dd. When contractor's field services are required (see 6.5.2).

6.3 Warranty. The successful bidder shall be required to guarantee the tractors and parts thereof against defective material and workmanship for a period of 12 months or 1500 operating hours, whichever occurs first, from date of acceptance of the equipment. On equipment used within the 50 States of the United States and the District of Columbia, the guarantee shall include the furnishing without cost to the Government, f.o.b. manufacturer's nearest dealer or branch, or to the original destination if desired by the Government, of new parts and assemblies to replace any that prove to be defective within

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five working days during the warranty period. In addition, when the Government elects to have the work performed by the contractor, the cost of labor involved in the replacement of the defective parts or assemblies shall be borne by the contractor. On equipment used outside the 50 States of the United States and the District of Columbia, the guarantee shall include the furnishing of new parts or assemblies to replace any returned to the contractor by the Government which prove to be defective. The replacement parts or assemblies shall be delivered by the contractor to the port of embarkation in the United States designated by the Government. The contractor shall not be required to bear the cost of the labor involved in correcting defects in equipment used outside the 50 States of the United States and the District of Columbia. The contractor shall further guarantee the tractor(s) conform to all federal laws and regulations governing safety, noise levels, and pollution which are required of the manufacturer for this type equipment at the date of solicitation.

6.4 Technical literature.

6.4.1 Manuals, operating and servicing. The successful bidder shall furnish one operator's manual, one lubricating chart, and one parts book with each tractor.

6.4.2 Illustrations and literature. If model data and drawings are desired, invitations for bids should specify that the bidder shall furnish with his bid two sets of illustrations or descriptive literature to assist the purchasing and using agencies in determining whether the tractors offered are adequate to perform the work specified (see 6.2).

6.5 Repair parts and service availability. If available service information is desired, invitations for bid should require data on repair parts and services.

6.5.1 Repair parts and service. As the continuous operation of the tractor contemplated by this specification is of utmost importance, it is necessary that the successful bidder be in a position to render prompt service and to furnish replacement parts. Accordingly, bidders shall indicate the extent of their ability to render prompt service by furnishing the location of their nearest dealer and branch office to the requisitioning activity where stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the parts book.

6.5.2 Contractor's field service. When specified (see 6.2), upon receipt of the equipment at the destination, the contractor shall furnish to the Government at no extra cost the services of a fully qualified, trained mechanic or serviceman to supervise and inspect the proper assembly and operation of the equipment. It is required that under his supervision and responsibility the tractor be completely adjusted, all equipment installed, completely lubricated with the grade of lubricant recommended by the manufacturer, battery fully charged, and made ready for continuous operation. The cooling system shall be checked for proper operating level and appropriate

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solution. In addition, the Government's operator shall be instructed in the proper operation and maintenance of the tractor. All materials required for the foregoing operations shall be furnished by the contractor. The contractor shall also furnish, when specified, a minimum of one free follow-up service call to be made by a competent, fully qualified, trained mechanic or serviceman. This service call shall be made either within 45 days after complete delivery of the equipment or after 30 days' usage, as designated after such delivery by the contracting officer or his duly authorized representative.

6.6 First article. When first article inspection is required, the item should be a production model. The first article should consist of one unit. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examination, test and approval of the first article test results and disposition of the first article.

6.7 Subject term (key word) listing.

Agricultural tractor
Tractor
Tractor, wheeled

6.8 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

MILITARY INTEREST:
Custodians
Army - ME
Air Force - 99

Review Activities:
Army - AT
DLA - CS

User Activities:
Navy - YD, MC

CIVIL AGENCY COORDINATING ACTIVITIES:
Interior - BLM
USDA - AFS
DOT - NHT
GSA - FSS
DCG

PREPARING ACTIVITY;
Army - ME

Project 2420-0028

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	KKK-T-640E	5 Dec 90

3. DOCUMENT TITLE
Tractor, Wheeled, Agricultural, (Gasoline-Eng-Drvn, DED or LP-GAS Engine Driven)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(703) 664-5717	354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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